SECTION 27 01 10

COMMUNICATIONS ROUGH-IN

PART 1 - GENERAL

1.1 SCOPE

A. Provide data, intercom, and security raceway, outlet boxes, junction boxes, terminal spaces, etc. as called for on the drawings and specified hereinafter. The raceway system shall be arranged to accommodate the installation of cable and equipment by others, unless otherwise specified hereafter, if specified hereafter it shall be the responsibility of the electrical contractor to furnish and install as part of the contract.

PART 2 - PRODUCTS

2.1 MATERIALS

A. Use intermediate metal conduit galvanized steel boxes above ground, or where mechanical strength or exposure to physical damage is required with PVC tubing used elsewhere.

B. Power, telephone, and other Security underground conduits where underground or concrete encased may be Schedule 40 PVC, transition to Schedule 80 PVC before emerging above grade or slab where exposed to physical damage.

C. Each length of conduit shall be stamped with name and trademark of manufacturer and approval of National Board of Fire Underwriters.

D. All conduit is to contain an equipment grounding conductor which is not illustrated on the drawings.

E. Minimum size conduit shall be ¾".

F. All conduit rough-in to have an installed pull tape or pull-wire for future use by others.

2.3 HANDHOLES AND BOXES

A. Description: Comply with SCTE 77.

1. Configuration: Units shall be designed for flush burial and have open bottom, unless otherwise indicated.

2. Cover: Weatherproof, secured by tamper-resistant locking devices and having structural load rating consistent with enclosure.

3. Cover Finish: Nonskid finish shall have a minimum coefficient of friction of 0.50.

4. Cover Legend: Molded lettering, "COMMUNICATTIONS" Or as indicated for each service.

5. Handholes 12 inches wide by 24 inches long and larger shall have factory-installed inserts for cable racks and pulling-in irons.

B. Polymer Concrete Handholes and Boxes with Polymer Concrete Cover: Molded of sand and aggregate, bound together with a polymer resin, and reinforced with steel or fiberglass or a combination of the two.

PART 3 – EXECUTION

3.1 STORAGE

A. Protect threads of galvanized rigid steel conduit and IMC during storage.

B. Stack conduit on blocking off ground to prevent the entry of foreign material.

3.2 INSTALLATION

A. Install factory conduit caps on stub-ups during construction. Swab trapped runs prior to pulling conductors. Take every precaution to prevent entry of water and foreign matter in conduit during construction.

B. Galvanized rigid steel conduit or "IMC" shall be terminated in threaded hubs or with double locknuts (Bondnut type) drawn tight and insulated conduit bushing.

C. Field cut conduit shall be cut square, reamed smooth and threaded properly and full. Paint field cut male threads with conductive and rust preventive compound. Cutting oil and debris shall be removed prior to installation.

D. PVC conduit shall be terminated with approved connectors and fittings. PVC conduit shall be heated and bent with manufacturer approved equipment and methods. Open flame or torch is not an acceptable means of heating. Field cuts shall be square and reamed smooth.

E. "EMT" conduit shall be terminated with couplings, connectors and fittings. Field-cut conduit shall be square and reamed smooth.

F. Conduit shall be installed and supported per National Electrical Code Article 342 (Intermediate Metal Conduit), Article 344 (Rigid Metal Conduit), Article 352 (Rigid Nonmetallic Conduit), Article 358 (Electrical Metallic Tubing) and Article 110 (Requirements for Electrical Installations).

G. Where exposed:
1. Organize the runs into groups and coordinate with other trades to avoid interference.
2. Arrangement shall be neat and orderly with runs parallel to structural elements. No diagonal runs will be allowed.
3. Supports shall be "Unistrut" with suitable clamps The Unistrut shall be supported from the building structures. Paint cut ends of Unistrut with rust prohibitor.

H. Underground conduits shall be located a minimum of 24" below finished grade. Locate a marker tape 12" below grade directly above the underground conduit.

3.3 SHOP DRAWINGS

A. Intermediate metal conduit.
   1. Meet or exceed Federal Specifications UL1242.
   2. Meet or exceed ANSI C680.6.

B. Polyvinyl chloride schedule 40.
   1. Meet or exceed Federal Specifications WC-1094A.
   2. Meet or exceed NEMA TC2.
   3. Meet or exceed UL 651.

C. Handholes and Pull Boxes
   1. Meets or exceeds specified on drawings.

D. Innerduct
   1. Meets or exceeds specified on drawings.

END OF SECTION